



## Audio-based snore detection using deep neural networks

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### ABSTRACT

**Background and Objective:** Snoring is a prevalent phenomenon. It may be benign, but can also be a symptom of obstructive sleep apnea (OSA) a prevalent sleep disorder. Accurate detection of snoring may help with screening and diagnosis of OSA.

**Methods:** We introduce a snore detection algorithm based on the combination of a convolutional neural network (CNN) and a recurrent neural network (RNN). We obtained audio recordings of 38 subjects referred to a clinical center for a sleep study. All subjects were recorded by a total of 5 microphones placed at strategic positions around the bed. The CNN was used to extract features from the sound spectrogram, while the RNN was used to process the sequential CNN output and to classify the audio events to snore and non-snore events. We also addressed the impact of microphone placement on the performance of the algorithm.

**Results:** The algorithm achieved an accuracy of  $95.3 \pm 0.5\%$ , a sensitivity of  $92.2 \pm 0.9\%$ , and a specificity of  $97.7 \pm 0.4\%$  over all microphones in snore detection on our data set including 18412 sound events. The best accuracy (95.9%) was observed from the microphone placed about 70 cm above the subject's head and the worst (94.4%) was observed from the microphone placed about 130 cm above the subject's head.

**Conclusion:** Our results suggest that our method detects snore events from audio recordings with high accuracy and that microphone placement does not have a major impact on detection performance.

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### 1. Introduction

Obstructive sleep apnea (OSA) is one of the most common sleep disorders, and is characterized by frequent partial or complete collapse of the upper airway during sleep. It affects 9% to 17% of the general adult population [1]. Studies indicate that OSA can increase risks of hypertension, cardiovascular disease, and stroke [2]. It leads to diminished quality of life and may cause problems, such as daytime sleepiness and, consequently, motor vehicle accidents [2].

Polysomnography (PSG) is the gold standard for diagnosing OSA, which requires overnight recordings of patients with multiple sensors in a sleep lab [3]. Although this method generates accurate results, it is associated with several difficulties, such as

long waiting lists, high nursing costs, supervision by a specialized sleep technician, and patient discomfort due to multiple sensors attached to the body [3,4]. Moreover, single night recordings can be suboptimal for some patient groups [5]. Therefore, there is a great need for the development of more efficient, cost-effective, and contactless methods for screening and diagnose OSA.

One of the earliest and most common symptoms of OSA is snoring. It occurs in 70% to 95% of OSA patients [4,6]. Snoring sounds are generated by the vibration of soft tissues in the upper airway, such as pharyngeal walls, soft palate, uvula, and tonsils, consecutive to a partial obstruction [7]. Detecting snore sounds can be the first step to explore a contactless OSA detection method. Questionnaires used to be the most frequent tool to study snoring, but are dependent on the regular presence of a bed partner, can be affected by their subjective impressions and it is difficult to validate their reliability [7]. Thus, a reliable and objective snore detection method could be advantageous.

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A number of studies of snore detection based on audio recordings from non-contact microphones have been carried out. Duckitt et al. [8] presented a system that uses Hidden Markov models (HMMs) on Mel-frequency cepstral coefficients (MFCCs) to identify snores from ambient acoustic data. They trained the HMMs over 1 hour recording of each subject and tested on 0.5 hour recording of each subject from 6 subjects and achieved sensitivity in the range of 82–89%. Cavusoglu et al. [9] classified snoring and non-snoring audio signals by using a robust linear regression model fitted on spectral features and they used a simple classification threshold to make the final decision. Audio recordings from 18 benign and 12 OSA snorers were trained and tested separately with half of the subjects used in a training set and the other half held out for testing. The sensitivity in the groups was 90.2% and 86.8% respectively. Azarbarzin and Moussavi [10] proposed an unsupervised snore detection algorithm that employs the fuzzy C-means (FCM) clustering on features from audio signals. They evaluated this algorithm on 23 OSA snorers and 7 simple snorers with unspecified evaluation rules and reported an overall sensitivity of 93.1%. Dafna et al. [11] developed a method that exploits an AdaBoost classifier on features extracted from enhanced audio signals to detect snoring and non-snoring acoustic events. They considered 67 subjects (25 for training and 42 for testing), under a rigorous evaluation protocol. The average accuracy reported was 98.2%. Swarnkar et al. [12] trained an artificial neural network (ANN) classifier on multiple audio features to classify snores and non-snores. The classifier used data from 21 subjects for training and achieved accuracy ranging between 86% and 89% on 13 testing subjects. Sun et al. [13] proposed a deep convolutional architecture trained with raw sound recordings to recognize snore segments. They included 10 subjects in the study and used 1-minute segments instead of snore events. An average precision of 81.82% was achieved, but the author did not specify their validation approach. Arsenali et al. [14] introduced a recurrent neural network (RNN) on selected MFCCs as input features to distinguish between snore and non-snore audio segments. Data from 20 subjects (a subset of the recordings included in this study) have been considered with 11 subjects used for training, 3 for validation, and 6 for testing. Their method achieved an accuracy of approximately 95% when evaluated over the test set. Shin and Cho [15] proposed a snoring detection technique for a smartphone application. They trained a quadratic classifier with formant related features extracted from audio recordings recorded by smartphones. They included 10 healthy subjects and used 10-fold cross-validation to validate the technique. This technique achieved an accuracy of 95.07%.

Although several of the mentioned studies yielded relatively accurate or at least promising results, they involved manual feature extraction, which requires not only domain-specific and signal processing knowledge, but also a lot of human effort. In this paper, we introduce a deep learning approach for snoring detection using audio signals. The network architecture primarily consists of three convolutional neural network (CNN) layers and one recurrent neural network (RNN) layer. The CNN layers are used to extract features automatically (without human effort on feature engineering) from two-dimensional (2D) representation of audio data [16,17]. Constant-Q transformation (CQT) has been chosen to convert an audio signal to its associated 2D representation due to its higher spectral resolution in the low frequency range, similarly to the Mel scale. It has been proposed by Youngberg and Boll [18] and later by Brown [19] and has been used for music processing [20–22]. The RNN layer is used to explore sequential characteristics of audio data, which has been widely used in other audio-based applications such as speech recognition [23,24] and natural language processing [25–27].

In addition, in the previous studies the microphone is almost always placed at one fixed position, which seems to be ad-hoc,

**Table 1**  
Subject demographic diagnostic information.

|                        |                               |
|------------------------|-------------------------------|
| Male/Female            | 28/10                         |
| Age                    | 54 ± 13 (range: 24–79)        |
| BMI                    | 29.1 ± 5.4 (range: 20.6–39.8) |
| AHI                    | 25.4 ± 19.3 (2.4–98.7)        |
| OSA diagnosis (yes/no) | 33/5                          |

and different from study to study (see table 6). There is a lack of consistency and recommendations for snore recording techniques [7,28,29]. The effect of microphone placements on recording of snoring is investigated by a limited number of studies. Ng et al. [30] evaluated the effects of the mouth-to-microphone distance on snore signal quality and patient comfort. This study suggested the optimal mouth-to-microphone distance to be 0.3 m. Herzog et al. [31] studied the relationship between microphone placement and the results of the frequency analysis of snoring sounds. They observed that air-coupled microphones have a wider frequency range than body contact microphones. However, it is not clear from their study how the microphone placement could affect snore detection performance.

In the present work, audio data collected by 5 microphones in a clinical setting (sleep lab) has been processed and analyzed. Snore detection performance has been compared with the aim of identifying the impact on snore detection accuracy. We also discuss the performance while taking into account the subject's body position to achieve a deeper understanding of the situation.

## 2. Materials and Methods

### 2.1. Data

#### 2.1.1. Subjects

The dataset used is a subset of the large SOMNIA database [32]. Based on availability of audio recordings, 38 subjects were included. All the subjects underwent routine PSG monitoring in the sleep lab at Kempenhaeghe Center for Sleep Medicine, Heeze, the Netherlands, between June 2017 and November 2017. No specific selection requirements were imposed in terms of gender, body mass index (BMI), and AHI. Subjects younger than 18 years and those treated with continuous positive airway pressure therapy were excluded. Table 1 summarizes demographic and diagnostic information of these subjects.

The SOMNIA study was reviewed by the medical ethical committee of the Maxima Medical Center (Eindhoven, the Netherlands). File no: N16.074 and W17.128). All participants provided written informed consent. All the studies met the ethical principles of the Declaration of Helsinki, the guidelines of Good Clinical Practice and the current legal requirements. The protocol for this data analysis study was approved by the Review Committee of Kempenhaeghe Center for Sleep Medicine and by the Philips Institutional Review Board (Internal Committee on Biomedical Experiments).

#### 2.1.2. Audio data recording

Two types of microphones (Earthworks M23 and Behringer ECM8000) were used for digital audio recording. Five of them were placed as indicated in Fig 1. They are tagged as M1–M5 with M1 and M2 being of Earthwork M23 type, and M3–M5 of Behringer ECM8000 type. The sampling frequency was set to 48 kHz and the sampling resolution to 24 bits (bit depth). As seen in the figure, M1 and M2 were placed above a subject's head with an approximate distance of 70 cm and 130 cm, respectively. M3 was located at the left side of the bed and M4 at the right side of the bed. M5 was placed on the bedside table to mimic smartphones in a home environment. The sleep institute is located in the country-side in a



**Fig. 1.** Microphone setup of sleep lab setting, where the microphone placements (M1-M5) are indicated in yellow. The height of the bed was between 65 cm and 85 cm and can be slightly adjusted by subjects. The bedside table has similar height as the bed. The distance between M3, M4 and the floor was 102 cm. Microphone M5 was placed on the bedside table around 100 cm away from subject's head (in a few cases this table could have been moved by subjects), the position of the bedside table depends on the specific room setup, and could be either on the left or on the right side of the subject (27 left, 11 right).

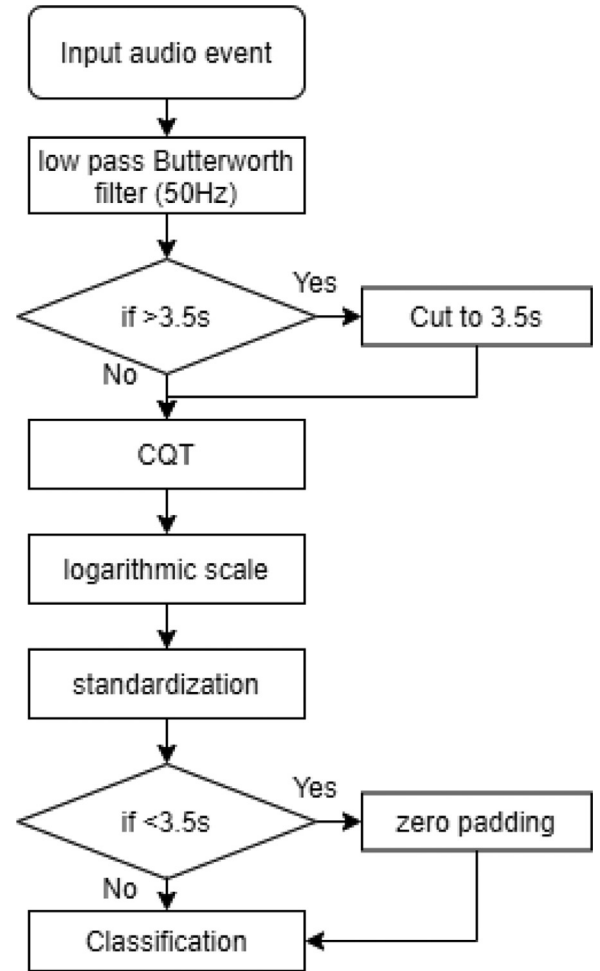
sparsely populated area, and the level of background noise in the respective lab rooms may be qualified as low during night.

### 2.1.3. Data annotation

Manual detection of sound events is a labor-intensive process. To speed this up, we use an automatic selection of sound intervals described by Arsenali et al. [14], followed by a manual annotation in terms of snore, non-snore, probably snore and probably non-snore events, done by an experienced sleep technician.

The automatic detection method by Arsenali et al. [14] has four steps. First, the raw signal is processed by an A-weighting filter that accounts for the relative loudness perceived by the human ear. Second, a rolling window with a duration of 0.1 s and a stride of 1 sample is used to estimate the power of the filtered signal. Third, a dual threshold is used to detect sound events from the estimated power of the filtered signal. The detection thresholds (i.e., low and high) are selected manually for each recording to account for the change in the noise floor caused by for example an air conditioning system. Finally, sound events with a duration of less than 0.2 s are excluded from further analysis. This procedure resulted in an average of 4756 sound events per subject (min and max of 1793 and 8509, respectively).

Next, manual annotations were facilitated by a graphical user interface that enables audio-visual inspection of automatically detected sound events and their surrounding environment was used to speed up the annotation process. Each event was labeled by one annotator with one of the following labels: snore, non-snore, probably snore, and probably non-snore. The two labels were used to indicate events that do not belong neither to clear snore nor to clear non-snore (i.e., they were used to label the unsure events). The annotator was instructed to focus on the recurrent snore patterns. This was done for two reasons. First, these patterns usually contain a similar number of snore and non-snore sound events, which results in a well-balanced dataset. Second, the annotation process is usually faster when annotating within a snore sequence than when annotating outside of this pattern. The aforementioned procedure resulted in a total of 20225 annotated sound events, including 7936 (39.2%) snore events (ranged from 25 to 439 per subject), 10476 (51.8%) non-snore events, and 1813 (9%) unsure events (annotated as probably snore and probably non-snore). Unsure events were excluded from this study because it was unclear whether those events were snore or heavy breathing.



**Fig. 2.** Pre-processing flowchart for snore detection.

## 2.2. Method

### 2.2.1. Data pre-processing and representation

Fig. 2 shows the main data processing steps. For each sound event, a low pass filter with a cutoff frequency of 50 Hz and an order of 3 was applied to its associated audio signal to remove the electrical power supply noise. To ensure that all inputs to the CNN layers have an equal length, we truncated the sound events longer than 3.5 s and used zero-padding for shorter events before processing. The length of 3.5 s was decided by inspecting the duration of all annotated sound events. Fig. 3 presents the distributions of the time length of all sound events, from which we can see that most of the sound events (>99%) were shorter than 3.5 s with a mean duration of 1.3 s. Thus, 3.5 s was selected to reduce unnecessary computation and keep as much information.

After cutting the events to 3.5s, the signal went through CQT, resulting its 2D spectrogram representation. This representation shows both the time and frequency domains and achieves input data reduction. In CQT, Q stands for the Q-factor which is the ratio of center frequency to bandwidth. The Q-factors of all frequency bins are equal. Here we briefly introduce the mathematical calculations of CQT and more details can be found elsewhere [18,19]. The Q-factor can be represented by

$$Q = \frac{f_k}{\Delta f_k} \quad (1)$$

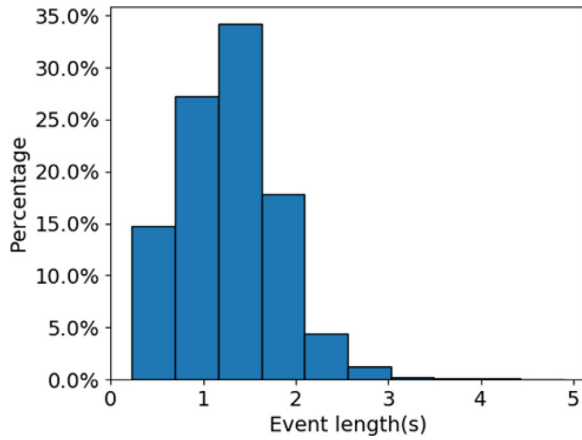


Fig. 3. Histogram of all sound event lengths.

where  $\Delta f_k$  is the bandwidth and  $f_k$  is the center frequency of  $k$ -th bin. The  $f_k$  values are obtained by

$$f_k = f_1 2^{\frac{k-1}{B}} \quad (2)$$

where  $f_1$  is the center frequency of the lowest frequency bin and  $B$  denotes the number of bins per octave. Based on these two equations, the CQT spectrogram has a higher frequency resolution in the low frequency range and a higher time resolution in the high frequency range. This important property makes CQT more suitable for snore signal processing than short-time Fourier transform on a linear frequency scale because snore sounds have more components in the low-frequency bands than in the high-frequency bands [28]. Additionally, the frequency scale of CQT is close to the human auditory system [20], which is similar to the well-known Mel-frequency scale, however, with even more details in the low frequency bands.

For a discrete time-domain signal  $x(n)$ , its CQT  $X(k, n)$  is defined by

$$X(k, n) = \sum_{j=n-\lfloor N_k/2 \rfloor}^{n+\lfloor N_k/2 \rfloor} x(j) a_k^*(j - n + N_k/2) \quad (3)$$

where  $\lfloor \cdot \rfloor$  denotes rounding towards negative infinity,  $a_k^*(n)$  is the complex conjugate of  $a_k(n)$ . The basis functions  $a_k(n)$  are complex-valued waveforms, which are defined as

$$a_k(n) = \frac{1}{N_k} w\left(\frac{n}{N_k}\right) \exp\left[-i2\pi n \frac{f_k}{f_s}\right] \quad (4)$$

where  $f_s$  is the sampling frequency,  $w(t)$  is a continuous window function (Hanning in this work) which is sampled at  $N_k \in \mathbb{R}$  in (3)-(4) denotes the window length.

CQT calculations were implemented using the Python library *LibROSA* (<https://librosa.github.io/librosa>) using the recursive sub-sampling method proposed by Youngberg and Boll [18]. A hop length of 512 samples and a bin number of 84 (with 12 bins per octave) were chosen when calculating CQT energies. The calculated spectral energies were then converted with a logarithmic scale (in decibels) to cope with the wide range of energy intensity. An example of an audio signal including 7 snore events and the corresponding CQT spectrogram is shown in Fig. 4. It can be observed that the dominant frequency components of snore events are mainly distributed over the low frequency range and CQT spectrograms provide an accurate representation.

Afterward, the spectral energies were normalized to have zero mean and unit standard deviation, and zero-padding was applied to ensure a length of 3.5 s. This determined the input samples submitted to the classification neural network.

## 2.2.2. Deep neural network architecture

As mentioned before, our deep neural network architecture was constructed with primarily three CNN layers and one RNN layer, as shown in Fig. 5. The architecture was designed based on experimenting on training loss and accuracy.

CNN layers were used to automatically extract time-frequency domain features from the CQT spectrogram without changing the shape of the input along the time axis. Square filters that are often used in CNN were changed into rectangular filters to obtain more features from frequency domain [33]. These filters have a length of 8 data points (pixels) along the frequency axis 2 along the time axis. For each CNN layer, 'ReLU' was used as the activation function, followed by a 2D max-pooling layer, and a dropout layer with a dropout rate of 0.5 to reduce overfitting.

With respect to RNN, Long Short-Term Memory (LSTM) model was chosen in our architecture because it is designed to characterize sequential data with long-range dependencies, and it is more effective than conventional RNN models for acoustic-related applications [34]. A dynamic LSTM layer was employed to process variable sequence lengths. Features obtained from CNN and the length of the spectrogram of the sound events along the time axis were together fed into the dynamic LSTM layer with 64 hidden units. After that, a fully connected layer with a 'softmax' activation function was connected to generate model output (i.e. detection of snore events).

During the training, the adaptive moment estimation algorithm (Adam) with a learning rate of 0.001 was applied as the optimizer and cross-entropy was used as the loss function. The batch size and epoch number were empirically set to 64 and 100, respectively. Those hyperparameters were decided experimentally.

## 2.2.3. Validation

We used 5-fold cross-validation to iteratively train and test our algorithm. The 38 subjects were divided into five groups with three groups of 8 subjects and two of 7 subjects. During each round of the cross-validation, four groups were used for training and the remaining one group was used for testing. This method ensures "subject-independent" modeling and evaluation where data from the same subject are not simultaneously present in both training, and testing groups. After running cross-validation, we got five sets of detection outputs. We applied a classification threshold of 0.5 on these outputs to obtain snore/non-snore classifications. Then we evaluated the performance of each subject. Afterward, all these sets of results were pooled together to evaluate the overall performance of the algorithm. Classification results were presented by confusion matrices, positive predictive value (PPV), negative predictive value (NPV), sensitivity, specificity, and accuracy. For a fair comparison, audio data recorded by different microphones were trained and validated separately.

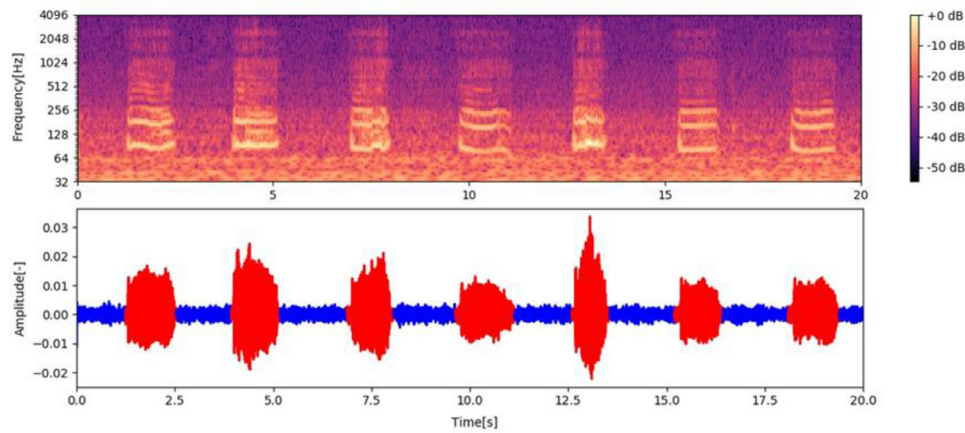
## 2.2.4. Microphone placement analysis

To study the influence of microphone placement on snore detection accuracy, the body position information of all snore events was extracted from the PSG data. There are 4 different body positions in total:

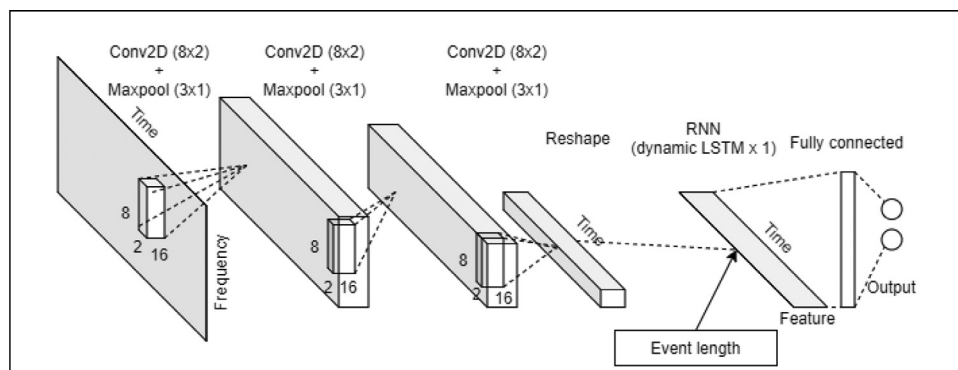
- 1 Sleeping on the right shoulder (body faces M3),
- 2 Sleeping on the back (supine, body faces M1 and M2),
- 3 Sleeping on the left shoulder (body faces M4),
- 4 Sleeping on the belly (prone).

The detection rates of snore events of each body position for each microphone were monitored and analyzed to evaluate the impact of microphone placement on classification performance. In order to make sure that one microphone only gets one body position facing it, we switched body position left and right for M5 when M5 is on the left side of the bed as M5 is not fixed in one position





**Fig. 4.** Example of an audio signal containing 7 snore events. The bottom part of the figure shows an example of raw signal after applying a Butterworth HP filter, where the red parts are snore events. The top part of the figure depicts the spectrogram based on CQT.



**Fig. 5.** The proposed deep neural network architecture for snore detection.

**Table 2**  
Confusion matrix of snore detection using different microphones (5-fold cross-validation).

|                |    |           | Ground Truth |           |
|----------------|----|-----------|--------------|-----------|
|                |    |           | Snore        | Non-snore |
| Classification | M1 | Snore     | 7351         | 178       |
|                |    | Non-snore | 585          | 10298     |
|                | M2 | Snore     | 7186         | 277       |
|                |    | Non-snore | 750          | 10199     |
|                | M3 | Snore     | 7297         | 179       |
|                |    | Non-snore | 639          | 10297     |
|                | M4 | Snore     | 7397         | 268       |
|                |    | Non-snore | 539          | 10208     |
|                | M5 | Snore     | 7358         | 302       |
|                |    | Non-snore | 578          | 10174     |

(27 right, 11 left). After this, the detection rate for M5 in the right body position stands for the detection rate when the body faces M5.

### 3. Results

Confusion matrix of the snore and non-snore classification was obtained from each microphone data stream over the sound annotations available for all 38 subjects. The 5 confusion matrices are presented in Table 2 from which the main performance indices of the classifier have been computed in terms of positive predictive value (PPV), negative predictive value (NPV), sensitivity, specificity, and accuracy. Table 3 reports these values in percentages for each microphone condition as the pooled values (overall performance). The pooled accuracy values are in the range between

94.4% and 95.9% and, although similar, they show interesting differences across microphones. When using the microphone positioned around 70 cm above the subject (M1), our method achieves the best overall performances in terms of PPV (97.6%), specificity (98.3%), and accuracy (95.9%). The second-best results were obtained with M3 and M4, which are placed at either left or right side of the subject at a fixed distance. M5 is placed further away on the bedside table while M2 located 130 cm above the subject leads to the worst performances for most metrics. By interpreting the pooled sensitivity and specificity values as detection rates of snore and non-snore sounds, respectively, it can be observed that non-snore events are better identified than snore events, with an average difference of 5.5%, a maximum of 6.9 % for M2 and a minimum of 4.2% for M4.

In Table 3, apart from the overall pooled performance, we also indicate the median and interquartile range (IQR) over each subject, for all microphones. In this way, the classifier performance can be assessed at the individual level. All the median values are higher than the pooled values with PPV in the range of 97.1% to 98.2%, NPV between 97.0% and 98.7%, sensitivity between 95.4% and 97.7%, specificity between 95.4% and 97.7%, and accuracy between 96.3% and 97.8%. They show a similar trend as the overall performance with M1 having a better performance, and M2 and M5 having a worse performance. Almost all IQRs are around 5% except sensitivity. Although the median sensitivity is much higher than the pooled sensitivity, the high IQRs (around 10%) show that quite a few subjects have many snore events misdetected. Using a Wilcoxon rank-sum test to test the significance of difference between M1 and other microphones, we found that only the specificity using M2 is significantly lower than using M1 ( $p < 0.05$ ).

**Table 3**

PPV, NPV, sensitivity, specificity, accuracy of each microphone for snore detection (5-fold cross-validation).

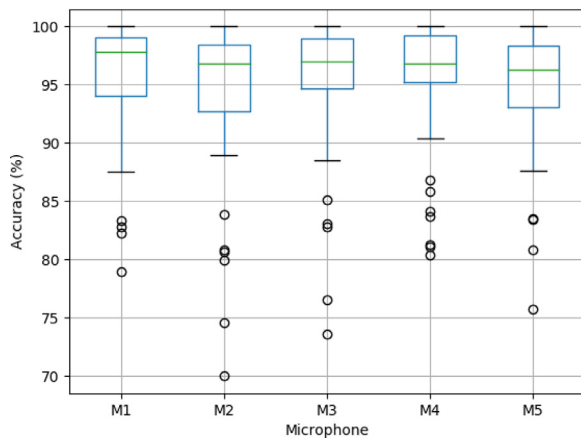
|                                  | M1 (%)            | M2 (%)            | M3 (%)            | M4 (%)           | M5 (%)            |
|----------------------------------|-------------------|-------------------|-------------------|------------------|-------------------|
| Positive Predictive Value (PPV)  | 97.6, 98.2 (3.9)  | 96.3, 97.3 (4.7)  | 97.6, 98.4 (5.5)  | 96.5, 97.3 (5.9) | 96.1, 97.1 (8.7)  |
| Negative Predictive Value (NPV)  | 94.6, 98.4 (7.2)  | 93.1, 97.8 (5.5)  | 94.2, 97.1 (6.1)  | 95.0, 98.7 (4.9) | 94.6, 97.0 (6.6)  |
| Sensitivity (True positive rate) | 92.6, 97.7 (10.8) | 90.5, 95.4 (11.1) | 91.9, 96.3 (12.4) | 93.2, 97.1 (9.4) | 92.7, 96.5 (10.2) |
| Specificity (True negative rate) | 98.3, 99.0 (1.9)  | 97.4, 98.1 (2.5)  | 98.2, 98.9 (3.0)  | 97.4, 98.7 (2.9) | 97.1, 98.4 (4.1)  |
| Accuracy                         | 95.9, 97.8 (5.0)  | 94.4, 96.9 (5.7)  | 95.6, 97.0 (4.2)  | 95.6, 96.8 (4.0) | 95.2, 96.3 (5.3)  |

Note: results are presented as pooled value, median (IQR).

Significance of difference was tested with Wilcoxon rank-sum test at  $p < 0.05$ .

\*The specificity using M1 is significantly higher than using M2.

Other comparison between microphones showed no significant difference.

**Fig. 6.** Boxplot of accuracy of all subjects with different microphones for snore detection.**Table 4**

Distribution of body positions for all snore events.

| Position                      | Snore events | Percentage |
|-------------------------------|--------------|------------|
| Right (body faces M3)         | 2797         | 35.2%      |
| Supine (body faces M1 and M2) | 3146         | 39.6%      |
| Left (body faces M4)          | 1971         | 24.8%      |
| Prone                         | 22           | 0.3%       |
| Total                         | 7936         | 100%       |

To further illustrate the performance at individual level, Fig. 6 shows the boxplot of accuracy for all subjects with all 5 microphones. We can observe that most subjects have a high accuracy, with some outliers having an accuracy below 80%. In general, M2 presents a lower accuracy than other microphones.

To analyze the relationship between microphone placement and performance, we first extracted the body position information from PSG data (in Table 4). Most snore events occurred in the supine and the right position in our dataset. Fewer snore events happened in the left position and very few in the prone position. Table 5 presents the snore events detection rates in each body position for each microphone. For snore events in the right position, M3 and M5 have higher detection rates. Regarding snore events in the left position, both M1 and M4 have higher detection rates, and M5 has the worst performance in this position. The detection rates in the lateral position are slightly better than the detection rates in the supine position for all microphones. M2 has the worst performance for detecting snore events in supine and right body positions of all microphones.

#### 4. Discussion

We developed an automatic snore detection algorithm from audio recordings based on a deep neural network model. The cur-

rent algorithm was developed and validated over 38 subjects. The overall accuracy of 5-fold cross-validation for recordings from all the microphones were above 94%. For the accuracy of each subject, no significant difference was found between M1, M3, M4, and M5, suggesting that the algorithm is robust to microphone placement and to between-subject differences.

Although the overall accuracy is high, 4.2% to 6.9% of all snore events were missed and a few subjects displayed a lower accuracy and sensitivity than average (Fig. 6). To find the reason, we have analyzed BMI, AHI, gender, and age, and found no statistical association between them and the performance. To further explain the missed snore detection, we conducted a thorough listening of all missed events, visually analyzed the CQT spectrograms and identified four main factors contributing to most misdetections:

- 1 Snore sounds produced during expiration (instead of during inspiration).
- 2 Low level of acoustic energy corresponding to weak loudness events.
- 3 Variability of snore sounds due to distinct places of pharyngeal constriction.
- 4 Inconsistencies among manual annotations of snore and non-snore events.

For the first factor, there is a small percentage of expiration snore sounds that have different sound characteristics from the major dominant inspiration snore sounds [35]. Their small prevalence makes it difficult for the algorithm to learn the characteristics of expiratory snores, which leads to the misdetection of these snore events. This problem was identified and addressed before by Dafna et al. [11]. Regarding the second factor, snore events with low-level acoustic energy have smaller signal to noise ratio and less clear spectrograms. Thus, these events are easily missed by our algorithm. When it comes to the third factor, the characteristics of snore sounds can be very diverse as they correspond to distinct places of partial constriction in the upper airways [36]. An example is depicted in Fig. 7, from which we can see CQT spectrograms of two snore events from different subjects with clearly distinct spectrograms. If one kind of snore sounds is not well represented in the training set, it could be missed during testing. This leads to the low accuracy of some specific subjects. The last factor concerns the accuracy of the annotations. Annotations were done by one single person without further verification or reconfirmation. Therefore, there is an element of subjectivity associated with the annotations and by definition not possible to guarantee that it is 100% accurate, which may contribute to the error rate. This factor is also mentioned by Dafna et al. [11] and has been solved by involving three annotators for the same dataset and applying a majority vote for the events without full agreement.

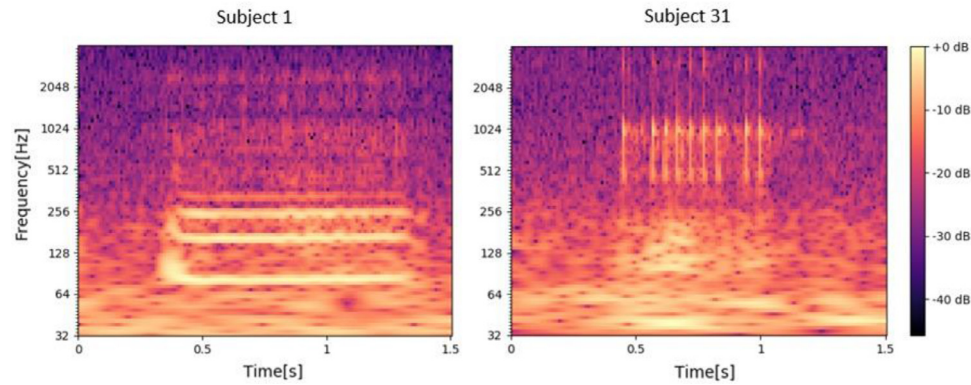
Apart from CQT spectrogram, Mel-scale spectrogram which is often used in speech recognition was also trained and tested with the same neural network model. It has been proved by Huzaiyah [37] that Mel-scale spectrogram consistently performs better than short-time Fourier transform (STFT) and CQT spectrogram for en-

**Table 5**

Detection rates of snore events (sensitivity) in different body positions for each microphone.

|    | Right (Body faces M3) | Supine (Body faces M1, M2) | Left (Body faces M4) | Prone |
|----|-----------------------|----------------------------|----------------------|-------|
| M1 | 94.5%                 | 88.1%                      | 97.0%                | 100%  |
| M2 | 92.8%                 | 85.2%                      | 95.9%                | 100%  |
| M3 | 95.7%                 | 86.9%                      | 94.5%                | 100%  |
| M4 | 93.4%                 | 90.8%                      | 96.7%                | 100%  |
| M5 | Body faces M5         |                            | Body against M5      |       |
|    | 94.7%*                | 90.6%                      | 93.0 %*              | 100%  |

\* These body positions do not necessarily correspond to true left and right sides (see section 2.2.4).

**Fig. 7.** Two different snore events. The left view shows a snore example of palatal vibrations (low F0 ~ 75 Hz), the right one is probably due to tongue or pharyngeal wall vibrations (F0 ~1 KHz).

vironmental sound classification using CNN. The accuracy of Mel-scaled spectrogram using data from M1 is 95.3% with a median of 96.9% and an IQR of 6.7%. This performance is slightly worse than CQT spectrogram. However, it would be hazardous to conclude that CQT spectrogram is consistently better than Mel-scaled spectrogram for snore detection because we only experimented one dataset and both CQT spectrogram and Mel-scale spectrogram can provide more details in low frequency range to some extent. To find the best two-dimensional representation of audio data, other time-frequency representations (e.g., STFT, wavelet transform) with a large dataset should be tested.

To compare our study with previous work, we summarized previous studies in Table 6. From this table, we can see that most methods require multiple features to be extracted from audio signals, which requires specific knowledge in feature extraction and selection algorithms. By contrast, our deep NN approach algorithm can obtain features automatically using CNN layers, which is the main novelty of this study. Although Sun et al. [13] used the raw audio signal, the PPV is much lower than ours. In addition, our algorithm outperforms most of the previous studies except the studies of Dafna et al. [11] and Shin and Cho [15]. But one disadvantage of the method of Dafna et al. [11] is that it requires 127 features. And for Shin and Cho [15], they only considered simple snorers. Arsenali et al. [14] used a subset of the same data in this study and evaluated their algorithm over 6 subjects with a snore detection accuracy of 95.4%, over the same subjects, our classifier achieves 97.5% accuracy.

Another contribution of this study concerns the influence of microphone placement on snore detection accuracy. This is not always addressed by the published studies which explore different microphone types and locations, as shown in Table 6. From Table 5, we can observe that the detection rate is slightly higher when the microphone faces the subject and slightly lower when the microphone faces the subject's back for lateral positions. In supine position, it is difficult to elaborate considering that when a subject sleeps on his back his head can still be turned to the left or right direction, facing other microphones. Nevertheless, we notice that

the detection rates in supine position are lower than the detection rates in other positions. After checking the misclassified events, we found that in our dataset, subjects with weak snore sounds of low acoustic energy are over-represented in supine position. This is unexpected since it is often observed that snoring is especially loud in supine position, due to the mechanical vibrations of velum or rear tongue resulting from a reduced upper pharyngeal section [38]. This is likely to be due to the selection of annotated sounds that are sampled over only a part of each night and do not represent the whole sleep period. This needs to be confirmed by further investigation.

Even with the different detection rates in different positions from different microphones, the overall performance for M1, M3, and M4 are similar (Table 2). Thus, we can conclude that the microphone placement does not have a major effect on the detection performance, evaluated under quiet external conditions (low background noise in sleep rooms). From Table 2 we can further observe that M2 has the worst overall performance. Since M2 has the largest distance (around 130 cm) from the subject, this suggests that distance may be a more important factor than microphone placement in terms of snore detection performance. In this study, we recommend hanging the microphone closely above the sleeper's head, as M1 has the best overall performance.

Although our algorithm achieves good performance for snore detection, there are limitations. The first one is the variability of the subjects. The characteristics vary substantially in terms of age, BMI, and AHI, which, together with the limited sample size, makes it difficult to compare the performance of this method on different groups. Furthermore, only a few snore events occur near apneas, which might be the reason that we could not find a correlation between these and the detection performance. The second factor is that we only use parts of the whole night recording. Without testing on full night recordings, it is hard to know how the method will generalize to the problems present with full night recordings in real world applications. The last limitation is the quiet background. Because of the low noise level of this dataset, it is un-

**Table 6**

Summary of previous studies on audio-based snore detection.

| Authors                      | Method                                         | Feature/Representation                                             | Microphone placement                                   | Dataset                                                                                                                 | Results                                                                                                          |
|------------------------------|------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Duckitt et al. [8]           | Hidden Markov models (HMMs)                    | MFCC-based feature vectors                                         | Bedside table                                          | 6 subjects, 1.5 h from each subject, 1 h for training, 0.5 h for testing                                                | Sensitivity, 82%–89%.                                                                                            |
| Cavusoglu et al. [9]         | Robust linear regression                       | Spectral features                                                  | 15cm over the patient's head                           | Full night recordings from 18 simple snorers and 12 OSA patients                                                        | Sensitivity and PPV, Simple snorer: 97.3%, 99.6%; OSA patients: Mixed, 90.2%, 98.7%; OSA patients: 86.8%, 93.8%. |
| Azarbarzin and Moussavi [10] | Unsupervised fuzzy C-means clustering          | Feature extraction using principal component analysis (PCA)        | Hang in the air                                        | A short period of the entire night recording from 23 OSA patients and 7 simple snorers                                  | Sensitivity and PPV, Simple snorer: 93.1%, 95.9% OSA: 91.6%, 97.2% Simple snorers: 96.4%, 93.5%                  |
| Dafna et al. [11]            | AdaBoost classifier                            | Time-related features and spectral-related features (127 features) | 1 meter above the head                                 | Full night recordings from 67 subjects (42 for validation)                                                              | Accuracy: 98.4%; Sensitivity: 98.1%; Specificity: 98.2%.                                                         |
| Swarnkar et al. [12]         | Artificial Neural Network                      | A set of features for speech analysis                              | Between 40cm to 70cm from the face of the patients     | Full night recordings from 34 subjects, 21 subjects for training, 13 subjects for testing                               | Accuracy: 86–89%; Sensitivity: 82–87%; Specificity: 87–89% PPV: 85–88%; NPV: 85–91%.                             |
| Sun et al. [13]              | SoundNet: a deep 1D convolutional architecture | Raw audio signal                                                   | A table 50cm away from the volunteer's head            | Part of full night recordings from 10 subjects                                                                          | Average PPV: 81.82%                                                                                              |
| Arsenali et al. [14]         | Recurrent neural network                       | MFCCs                                                              | Approximately 70 cm from the top end of the bed (M1)   | Part of full night recordings from 20 subjects (11 for training, 3 for validation and 6 for testing)                    | Accuracy: 95%; Sensitivity: 92%; Specificity: 98%.                                                               |
| Shin and Cho [15]            | Quadratic classifier                           | Features related to formant positioning and magnitude ratios       | Upper side of the shoulder, within arm's reach (<50cm) | Full night recordings after pre-selection from 18 subjects without preexisting respiratory diseases or sleep disorders. | Accuracy: 95.07%; Sensitivity: 98.58%; Specificity: 94.62%.                                                      |
| This work                    | CNN + RNN                                      | CQT spectrogram                                                    | Approximately 70 cm from the top end of the bed (M1)   | Part of full night recordings from 38 subjects                                                                          | Accuracy: $95.3 \pm 0.5\%$ ; Sensitivity: $92.2 \pm 0.9\%$ ; Specificity of $97.7 \pm 0.4\%$                     |

clear how the performance will hold when using recordings with a higher noise level.

## 5. Conclusion and future work

We propose a deep neural network model to classify snore and non-snore audio events from audio signals. The designed deep learning model does not use any manually engineered features and was trained and tested over audio data of 38 subjects from 5 microphones with different placements. We obtained robust performance for all microphone positions. This method outperformed most methods in previous studies. In addition, we analyzed the influence of microphone placements on the performance of this method and found that distance is more important than position.

Future work could proceed along different directions. The first one could be to further improve the accuracy of the current snore detection algorithm. This could be achieved by separating expiration from inspiration snore sounds, adding more events from different subjects to improve the variety of training samples, increasing the signal to noise ratio using noise reduction techniques, combining signals from different microphones to increase the coverage of high-quality signal and further checking the validity of annotations. Another direction could be to explore how can snore detection help with an OSA detection task. This study can be the first step of OSA detection as further analyzing acoustic properties of snoring may yield features diagnostic for the condition [39]. Apart from acoustic properties, other properties of snoring (e.g. the percentage of time snoring during sleep time) could also possibly be metrics for apnea analysis. In addition, some studies have shown differences between snores produced immediately after an apnea and other snore events [40,41]. Therefore, we could expand the current algorithm to separate those snore sounds to potentially help estimate AHI (e.g., [42]).

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## Ethics approval

Maxima Medical Center (Eindhoven, the Netherlands, File no: N16.074).

## Declaration of Competing Interest

P.F. and X.L. are employed by Philips Research. The employer had no influence on the study and on the decision to publish.

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